

Dr. Divyesh A. Mistry

Computational Materials Scientist

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Professional Summary

Computational materials scientist with over a decade of experience driving materials innovation through atomistic simulation, multiscale modeling, and high-performance computing. Proven ability to translate complex physical mechanisms into predictive computational tools for aerospace, defense, and maritime applications. Expertise spans molecular dynamics (LAMMPS), discrete dislocation dynamics (DDD), and finite element analysis, underpinned by automated Python workflows for simulation management, data pipelines, and publication-quality visualization on HPC clusters. Research portfolio covers Ni-based superalloys, tungsten alloys, and polymeric adhesive systems, with direct relevance to structural integrity, fracture prediction, and materials qualification.

Technical Skills

Simulation & Modeling: Molecular dynamics (MD), discrete dislocation dynamics (DDD), finite element analysis (FEA/FEM), multiscale modeling, constitutive modeling, fracture mechanics, plasticity, microstructure evolution.

Simulation Software: LAMMPS, OVITO, AtomsK, ABAQUS, ANSYS, COMSOL, NEPER, ParaView, MATLAB.

Programming & Scientific Python: Python (NumPy, Pandas, SciPy, Matplotlib), C++, Fortran, Bash, workflow automation, large-scale data analysis, Git/GitHub, L^AT_EX.

High-Performance Computing (HPC): Linux/Unix, MPI/OpenMP parallel computing, HPC cluster management, batch scheduling, multi-million-atom simulations.

Professional Experience

Postdoctoral Researcher

University of Seville

Aug 2026–Present

Seville, Spain

- Develop molecular-dynamics and multiscale workflows for cross-linked epoxy and polymer adhesive systems, linking network chemistry and crosslink density to thermomechanical response, glass-transition temperature, shear strength, and failure.
- Design reproducible LAMMPS protocols for curing, equilibration, cooling, and deformation; evaluate thermostat, strain-rate, and finite-size effects to improve verification and physical interpretation.
- Build Python pipelines for replica-based statistical analysis, stress-strain processing, property extraction, uncertainty assessment, and automated visualization of large simulation campaigns on HPC systems.
- Connect atomistic shear response with cohesive-zone and continuum descriptions of adhesive joints to support strength and life-prediction models across length scales.
- **Project:** *Multiscale modeling of interfacial delamination of steel/adhesive interfaces by aqueous corrosion*, funded by the U.S. Office of Naval Research.
- **Advisors:** Prof. María del Pilar Ariza Moreno and Prof. Michael Ortiz.

Postdoctoral Fellow / Visiting Researcher

Merrimack College, Department of Mechanical and Electrical Engineering

Apr 2026–Jul 2026

North Andover, MA, USA

- Performed large-scale MD simulations of temperature-dependent crack-tip deformation and the ductile-to-brittle transition in tungsten and W-Re alloys.
- Developed automated LAMMPS and Python workflows for stress-strain analysis, fracture-energy calculations, defect characterization, dislocation extraction, and trajectory processing on HPC clusters.
- Integrated atomistic fracture mechanisms with a 3D polycrystalline DDD framework to study Frank-Read source activation, dislocation multiplication, grain-boundary interactions, and plastic shielding.
- Mentored student research and contributed to manuscript development and multiscale simulation planning.

Junior Research Fellow, DRDO-Sponsored Project

Indian Institute of Technology Bombay

Jan 2025–Dec 2025

Mumbai, India (concurrent with final thesis defense)

- Investigated constitutive behavior and high-strain-rate response of shape-memory-alloy composites for aerospace and defense applications.
- Conducted Split Hopkinson Pressure Bar experiments and used experimental data to calibrate and validate numerical constitutive models.
- Supported project execution, data analysis, technical reporting, and mentoring of junior researchers.

PhD Researcher, Aerospace Engineering

Indian Institute of Technology Bombay

Jan 2019–Dec 2025

Mumbai, India

- Developed an atomistically informed multiscale framework for dislocation-precipitate-interface interactions in powder-metallurgy Ni-based superalloys.
- Executed multi-million-atom LAMMPS simulations to quantify generalized stacking-fault energies, dislocation mobility, precipitate shearing, critical resolved shear stress, and temperature-dependent mechanisms.
- Established size-dependent power laws for paired edge dislocations interacting with γ' precipitates and transferred atomistic drag and obstacle-strength parameters into DDD models.
- Developed polycrystalline DDD/finite-element models of prior particle boundaries, quantifying hardening, pinned

dislocation density, geometrically necessary dislocations, and residual-stress retention.

- Created parallel and automated Python/C++/Fortran workflows for simulation setup, defect tracking, post-processing, verification, parameter studies, and publication-quality visualization.
- Published first-author research, presented at international conferences, and collaborated with computational and experimental researchers.

Graduate Teaching Assistant

Indian Institute of Technology Bombay

Jan 2019–Dec 2023

Mumbai, India

- Supported Continuum Mechanics, Data Analysis, and Multiscale Modeling courses; developed computational exercises and guided undergraduate and postgraduate students.

Assistant Professor, Mechanical Engineering

CMR Institute of Technology

Jun 2016–Dec 2018

Bengaluru, India

- Taught Finite Element Methods, Experimental Stress Analysis, and Solid Mechanics; developed laboratory modules using ANSYS, MATLAB, and Python.

Graduate Teaching Assistant and MTech Researcher

Indian Institute of Technology Kharagpur

Jun 2014–Jul 2016

Kharagpur, India

- Developed and validated ABAQUS and ANSYS models for structural mechanics, smart composites, and vibration; supported laboratories in FEM, vibration analysis, and tribology.

Education

PhD, Aerospace Engineering, Indian Institute of Technology Bombay 2019–2025

Thesis: *Multiscale Modeling of Prior Particle Boundaries in Nickel-Based Superalloys*; CGPA: 9.36/10

Advisors: Prof. P. J. Guruprasad and Prof. Amuthan A. Ramabathiran

MTech, Mechanical Engineering (Mechanical System Design), IIT Kharagpur 2014–2016

Thesis: *A Mesh-Free Model for Static Analysis of Smart Composite Beams*; CGPA: 8.85/10

BE, Aeronautical Engineering (Aero Mechanical), Aeronautical Society of India 2013

Publications

Peer-Reviewed Journal Articles

1. **Mistry, D. A.**, Tak, T. N., Sankarasubramanian, R., and Guruprasad, P. J. “A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-Based Superalloys.” *Journal of Engineering Materials and Technology* (2026). DOI.
2. **Mistry, D. A.** and Ramabathiran, A. A. “Size-Dependent Power Laws for Edge Dislocations in Nickel Superalloys: A Molecular Dynamics Study.” *Computational Materials Science* 259, 114122 (2025). DOI.

Manuscripts in Preparation

1. **Mistry, D. A.** and Mahata, A. “Crack-Tip Plasticity and Ductile-to-Brittle Transition in W and W–Re Systems through Atomistic Fracture and Mesoscale Dislocation Dynamics.” Manuscript in preparation (2026).
2. **Mistry, D. A.**, Tak, T. N., and Guruprasad, P. J. “Polycrystalline Discrete Dislocation Dynamics of Ni-Based Superalloys: Interactions with Prior Particle Boundaries and Second Phases Using Atomistically Informed Inputs.” Manuscript in preparation.

Conference Contributions

1. EMI 2026, Boulder, USA: *Multiscale Modeling of Dislocation–Precipitate–PPB Interactions in Powder-Metallurgy Nickel Superalloys*.
2. IMPLAST 2025, IIT Roorkee, India: *Hierarchical Multiscale Modeling of Plasticity in Ni-Based Superalloys*.
3. COMPLAS 2023, Barcelona, Spain: *An Atomistically Informed DDD Study of Prior Particle Boundaries in Ni-Based Superalloys*.

Awards, Grants & Professional Memberships

- **Institution of Eminence Travel Grant**, IIT Bombay (2025) — funded attendance at TMS 2025, USA.
- **IIT Bombay International Travel Grant** (2023) — funded attendance at COMPLAS 2023, Spain.
- **GATE All India Rank: 69** (2014) — Graduate Aptitude Test in Engineering; MHRD fellowships for doctoral and master’s study.
- **Member**, The Minerals, Metals & Materials Society (TMS); **Associate Member**, Aeronautical Society of India (AeSI).
- Certifications: Using Python for Research (HarvardX); The Unix Workbench (Johns Hopkins); Material Behavior (Georgia Tech); Engineering Simulations (CornellX).

Additional Information

Research interests: computational materials science, multiscale mechanics, atomistic simulation, deformation and fracture, additive-manufacturing materials, polymer adhesives, and data-driven materials modeling. Academic references available upon request.